A Dirac Fermion in a Scalar Mass Growing Linearly with the Cylindrical Radius:

Channel Coupling, Spectral Bounds, and Numerical Spectrum

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Abstract

We study a Dirac fermion in 3+1 dimensions whose scalar mass grows linearly with the cylindrical radius, $M(\rho) = \kappa\rho$, the fermionic counterpart of a bosonic model with $m^2 \to m^2 + \kappa^2\rho^2$ studied recently. The axial momentum k separates exactly, $H(k)^2 = k^2 + H(0)^2$, because α^3 anticommutes with every matrix in the transverse Hamiltonian. Squaring $H(0)$ gives, in each angular-momentum channel l, a four-component radial operator K_l that splits into two unitarily equivalent two-component blocks, K_{I} and K_{II}. The block K_{I} couples two two-dimensional oscillator operators with centrifugal terms l^2 and $(l+1)^2$, through an off-diagonal matrix of magnitude κ. The coupling does not commute with the oscillator part, since the centrifugal terms differ by $(2l+1)/\rho^2 \neq 0$ for every integer l, and no constant change of component basis decouples the two oscillators. This distinguishes the problem from the one-dimensional linear scalar mass, which is solvable on each half-line, and from the bosonic and Dirac-oscillator problems, where the direct reduction to Kummer's equation is available. We show that the radial Dirac operator is essentially self-adjoint with $H(k)^2 = k^2 + K_l$ and, using the min–max principle, that K_{I} has a purely discrete spectrum with $\kappa(2n + 2m_l + 1) \leq \mu_n(l) \leq \kappa(2n + 2m_l + 3)$, where $m_l = \min(|l|, |l+1|)$ and $\mu_n(l)$ are the ordered eigenvalues of K_{I}. At fixed k the radial Dirac spectrum is $E = \pm\sqrt{k^2 + \mu_n(l)}$ with equal multiplicities, so the full Hamiltonian has continuous subbands separated by a gap $|E| < E_* = \sqrt{\mu_0(0)}$. Here $\mu_0(0)$ is the lowest eigenvalue of K_{I} in the channel $l = 0$, degenerate with the corresponding lowest eigenvalue for $l = -1$: the mass profile confines the fermion transversely but not along the axis. Matched shooting and first- and second-order Laguerre-basis calculations give for $\kappa = 1$ and $l = 0$ the levels $E_0 = E_* = 1.2810688331$, $E_1 = 2.0517799301$, $E_2 = 2.4146610347, \ldots$; numerically, the deviations $\mu_n - 2\kappa(n+1)$ alternate in sign and decrease in magnitude for all 45 levels computed (an observation, not a theorem). No closed form is found, and none is claimed to be impossible.

1 Introduction

A position-dependent mass enters the Dirac equation through the Lorentz-scalar term $\beta M(\boldsymbol{r})$. The scalar and vector (time-component) couplings behave differently at large distance. For the pure vector confining potentials studied by Giachetti and Sorace, including linear and harmonic examples, the would-be bound states appear as metastable resonances rather than normalizable bound states [2]. By contrast, a Lorentz-scalar mass term that grows at infinity can confine motion in the directions in which it grows. For the model studied here, which depends on the transverse coordinates only, this means transverse confinement, as we shall prove. Dirac equations with scalar potentials have a long history [3], including confining scalar-plus-vector combinations [4] and spherically symmetric linear scalar potentials used in quark-confinement models, studied numerically and, for special parameter relations, analytically [5, 6]. Position-dependent masses have been treated for the Dirac equation [8] and, in plane polar coordinates, for the Schrödinger equation [9, 10].

A recent study shows a similar mass growth this study will show, but that of the Casimir effect for a massive scalar field whose squared mass grows quadratically with the cylindrical radius, $m^2 \to m^2+\alpha^2\rho^2$ [1]. For that bosonic model the transverse operator is a two-dimensional harmonic oscillator and everything is known in closed form: with $\kappa = \alpha$ the squared frequencies are $\omega^2 = k^2 + m^2 + 2\kappa(2n + |l| + 1)$. Among possible extensions, Ref. [1] mentions analogous position-dependent-mass models for fermionic and gauge fields. The most direct fermionic analog is a Dirac fermion with scalar mass $M(\rho) = \kappa\rho$, for which $M(\rho)^2 = \kappa^2\rho^2$ reproduces the position-dependent quadratic term of the bosonic effective squared mass in the zero-bare-mass case.

Two neighboring problems have known analytic treatments. In one dimension, a linear scalar mass $M = m + g|x|$ reduces on each half-line to the parabolic-cylinder equation; the allowed energies are fixed by a transcendental matching condition at the kink [7]. In the plane, the Dirac oscillator, a non-minimal coupling linear in the coordinates, has a closed-form spectrum of Laguerre type [11, 12]. We are not aware of a previous treatment of the cylindrical linear scalar mass, and we show that its structure differs from both.

Results. (i) The axial momentum separates exactly, $H(k)^2 = k^2 + H(0)^2$ (Sec. 2). (ii) Squaring $H(0)$ gives, in every angular-momentum channel, a four-component radial operator K_l that splits into two unitarily equivalent two-component blocks. The block K_{I} couples two oscillators with different centrifugal terms through a constant matrix (Sec. 3). The coupling does not commute with the oscillator part in any channel, so no constant change of component basis decouples it, and the direct reduction to Kummer's equation available for the bosonic and Dirac-oscillator problems does not apply (Sec. 4). (iii) We show that the radial Dirac operator is self-adjoint with square $k^2 + K_l$, prove with the min–max principle [13] that the spectrum of K_{I} is discrete, bound every level within an interval of width 2κ, and show that at fixed k the Dirac spectrum is $\pm\sqrt{k^2 + \mu_n(l)}$, where $\mu_n(l)$ are the ordered eigenvalues of K_{I}. Because the axial momentum is not confined, the full Hamiltonian has continuous subbands with a gap (Sec. 5). (iv) We compute the spectrum with three numerical formulations (matched shooting and two Laguerre-basis calculations), tabulate the levels, and report regularities that go beyond the proven bounds (Sec. 6).

Scope. We do not include a bare mass and we do not claim that a closed form is impossible: our statement is that the structure producing closed forms in the neighboring problems is absent here. Units are $\hbar = c = 1$; $\kappa > 0$ has dimension (mass)^2, so that $\kappa = 1$ fixes the unit of energy to be $\sqrt{\kappa}$.

2 Model and exact reduction

2.1 Hamiltonian and radial equations

We use the Dirac representation, $\alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}$ and $\beta = \mathrm{diag}(1, 1, -1, -1)$, and study

$$H = \boldsymbol{\alpha} \cdot \boldsymbol{p} + \beta M(\rho), \qquad M(\rho) = \kappa\rho, \qquad \rho = \sqrt{x^2 + y^2}. \tag{1}$$

The fermion is massless on the axis. H commutes with p_z and with $J_z = -\mathrm{i}\partial_\varphi + \Sigma_z/2$, $\Sigma_z = \mathrm{diag}(\sigma_z, \sigma_z)$. Simultaneous eigenfunctions with $p_z = k$ and $J_z = l + \frac{1}{2}$, $l \in \mathbb{Z}$, have the form

$$\psi = \mathrm{e}^{\mathrm{i}kz} \begin{pmatrix} \mathrm{e}^{\mathrm{i}l\varphi} f_1(\rho) \\ \mathrm{e}^{\mathrm{i}(l+1)\varphi} f_2(\rho) \\ \mathrm{e}^{\mathrm{i}l\varphi} h_1(\rho) \\ \mathrm{e}^{\mathrm{i}(l+1)\varphi} h_2(\rho) \end{pmatrix}, \tag{2}$$

and $H\psi = E\psi$ becomes

$$(E - \kappa\rho)\, f_1 = k\, h_1 - \mathrm{i}\Big(h_2' + \frac{l+1}{\rho} h_2\Big), \tag{3a}$$

$$(E - \kappa\rho)\, f_2 = -k\, h_2 - \mathrm{i}\Big(h_1' - \frac{l}{\rho} h_1\Big), \tag{3b}$$

$$(E + \kappa\rho)\, h_1 = k\, f_1 - \mathrm{i}\Big(f_2' + \frac{l+1}{\rho} f_2\Big), \tag{3c}$$

$$(E + \kappa\rho)\, h_2 = -k\, f_2 - \mathrm{i}\Big(f_1' - \frac{l}{\rho} f_1\Big). \tag{3d}$$

Near $\rho = 0$ the derivative terms dominate. For each integer l the system then has two solutions that are square integrable with respect to $\rho\,\mathrm{d}\rho$ and two that are not; the latter behave as $\rho^{-|l+1/2|-1/2}$. At large ρ the solutions behave as $\mathrm{e}^{\pm\kappa\rho^2/2}$, of which only one is square integrable. Hence the radial operator is limit-point at both ends (Lemma 3), and no boundary condition beyond normalizability is needed: the physical solutions are the regular ones, with $f_1, h_1 \sim \rho^l$ for $l \geq 0$ and $f_2, h_2 \sim \rho^{-l-1}$ for $l \leq -1$.

2.2 Separation of the axial momentum

Lemma 1. *Write $H(k) = H(0) + k\alpha^3$ with $H(0) = \alpha^1 p_x + \alpha^2 p_y + \beta M(\rho)$. Then $\{\alpha^3, H(0)\} = 0$ and*

$$H(k)^2 = H(0)^2 + k^2. \tag{4}$$

Proof. α^3 anticommutes with α^1, α^2 and β, and commutes with the scalar functions p_x, p_y, $M(\rho)$; hence $\{\alpha^3, H(0)\} = 0$. With $(\alpha^3)^2 = 1$, $H(k)^2 = H(0)^2 + k\{H(0), \alpha^3\} + k^2 = H(0)^2 + k^2$. □

Equation (4) reduces everything to the k-independent operator $H(0)^2$, which is studied in Secs. 3–5.

2.3 Which couplings separate

The only property used in Lemma 1 is that every matrix multiplying a transverse operator in $H(0)$ anticommutes with α^3. This holds for minimal coupling to a gauge field, $\boldsymbol{\alpha} \cdot (\boldsymbol{p} - e\boldsymbol{A}_\perp)$, and for the scalar mass $\beta M(\rho)$. It fails for non-minimal couplings built from the matrices $\alpha^i\beta$, which commute with α^3. For the planar Dirac oscillator with a free axial direction, $H = \alpha^1\pi_x + \alpha^2\pi_y + k\alpha^3 + \beta m_0$, $\pi_j = p_j - \mathrm{i}m_0\omega\beta x_j$, one finds

$$H(k)^2 = H(0)^2 + k^2 - 2\mathrm{i}\, m_0\omega\, k\, \alpha^3\big(\alpha^1\beta\, x + \alpha^2\beta\, y\big), \tag{5}$$

and the cross term does not vanish. Additivity of E^2 in k^2 is then not a consequence of the Clifford algebra alone. The nonzero cross term shows that the simple additive separation in Eq. (4) is not a generic consequence of an axial momentum term. The scalar mass of this paper belongs to the class where the separation is exact.

3 The squared operator and channel coupling

3.1 Squaring

Since $\{\alpha^i, \alpha^j\} = 2\delta^{ij}$ and $\{\alpha^i, \beta\} = 0$,

$$\begin{aligned} H(0)^2 &= \boldsymbol{p}_\perp^2 + M^2 + \alpha^i\beta\,(p_i M - M p_i) \\ &= \boldsymbol{p}_\perp^2 + M^2 - \mathrm{i}\,(\partial_i M)\,\alpha^i\beta, \qquad i = 1, 2. \end{aligned} \tag{6}$$

For $M = \kappa\rho$ one has $\partial_i M = \kappa\,\hat{\rho}_i$, so that

$$H(0)^2 = \boldsymbol{p}_\perp^2 + \kappa^2\rho^2 + \kappa\,\Xi(\varphi), \tag{7}$$

$$\Xi(\varphi) = -\mathrm{i}\big(\cos\varphi\,\alpha^1\beta + \sin\varphi\,\alpha^2\beta\big). \tag{8}$$

The matrices $\alpha^1\beta$ and $\alpha^2\beta$ are anti-Hermitian, square to -1 and anticommute; hence Ξ is Hermitian with $\Xi^2 = 1$ and $\operatorname{tr}\Xi = 0$, so its eigenvalues are $-1, -1, +1, +1$ at every point. In particular $\|\Xi\| = 1$ as a multiplication operator.

3.2 Radial form and block structure

On the ansatz (2), Ξ acts as

$$\Xi:\ (f_1, f_2, h_1, h_2)\ \mapsto\ (\mathrm{i}h_2,\ \mathrm{i}h_1,\ -\mathrm{i}f_2,\ -\mathrm{i}f_1), \tag{9}$$

with the angular phases unchanged. Let $L_m = -\partial_\rho^2 - \rho^{-1}\partial_\rho + m^2/\rho^2$. Squaring the radial operator of Eqs. (3) gives, for all l and k,

$$\begin{aligned} (\text{radial } H)^2 &= k^2 + K_l, \qquad (10)\\ K_l &= \operatorname{diag}(L_l, L_{l+1}, L_l, L_{l+1}) \\ &\quad + \kappa^2\rho^2 + \kappa\,\Xi. \qquad (11) \end{aligned}$$

K_l is independent of k and splits into two 2×2 blocks. On $(f_1, h_2)^T$ and $(h_1, f_2)^T$,

$$K_{\mathrm{I}} = \begin{pmatrix} L_l + \kappa^2\rho^2 & \mathrm{i}\kappa \\ -\mathrm{i}\kappa & L_{l+1} + \kappa^2\rho^2 \end{pmatrix}, \qquad K_{\mathrm{II}} = \sigma_z K_{\mathrm{I}} \sigma_z. \tag{12}$$

The blocks are unitarily equivalent, so the spectrum of K_l is that of K_{I}, each eigenvalue occurring twice. We write $K_{\mathrm{I}} = A + \kappa\,\Xi_{\mathrm{I}}$, with $A = \operatorname{diag}(L_l + \kappa^2\rho^2,\ L_{l+1} + \kappa^2\rho^2)$ the oscillator part and $\Xi_{\mathrm{I}} = \begin{pmatrix} 0 & \mathrm{i} \\ -\mathrm{i} & 0 \end{pmatrix} = -\sigma_y$ the coupling.

3.3 Non-commutativity and the one-dimensional comparison

The two diagonal entries of A differ only through the centrifugal terms, $L_l - L_{l+1} = -(2l+1)/\rho^2$, and therefore, as an identity of differential operators on $C_c^\infty((0,\infty))^2$,

$$[A, \Xi_{\mathrm{I}}] = -\mathrm{i}\,\frac{2l+1}{\rho^2}\,\sigma_x \neq 0 \qquad \text{for every integer } l \tag{13}$$

The operator Ξ_{I} does not map the domain $D(A)$ into itself when $|l| \neq |l+1|$, because it moves a component with the regular behavior of one sector into the other; Eq. (13) is therefore a statement about test functions and not about all of $D(A)$. Its structural content is the following. Write $A = c\,\mathbb{1} - \frac{2l+1}{2\rho^2}\sigma_z$ with the scalar operator $c = \frac{1}{2}(L_l + L_{l+1}) + \kappa^2\rho^2$. Then

$$K_{\mathrm{I}} = c\,\mathbb{1} + \boldsymbol{n}(\rho)\cdot\boldsymbol{\sigma}, \qquad \boldsymbol{n}(\rho) = \Big(0,\ -\kappa,\ -\frac{2l+1}{2\rho^2}\Big), \tag{14}$$

and the direction of $\boldsymbol{n}(\rho)$ changes with ρ because $2l+1 \neq 0$. A constant unitary 2×2 matrix rotates all Pauli vectors rigidly, so no constant change of component basis makes K_{I} diagonal for all ρ; a ρ-dependent rotation would in general generate off-diagonal terms from the derivatives in c. For integer l the factor $2l+1$ is odd and never vanishes, so this obstruction is present in every channel. It does not exclude other ways of solving the problem.

In one dimension the corresponding statement is different. With $\alpha = \sigma_y$, $\beta = \sigma_x$ and $M = m + g|x|$ the same calculation gives $H^2 = p^2 + M^2 - M'\sigma_z$ with the *constant* matrix σ_z, which commutes with $p^2 + M^2$. On each half-line, where $M' = \pm g$ is constant, the two spinor components decouple into scalar equations with shifted harmonic-oscillator potentials $p^2 + (m + g|x|)^2 \mp g$. The scalar equation reduces to a parabolic-cylinder equation on each half-line; the allowed energies are selected by matching the solutions across the kink at $x = 0$ [7]. In the plane the analog of σ_z is $\Xi(\varphi)$, which depends on the angle; in a fixed channel it becomes a constant matrix, but then the oscillator part acquires two different centrifugal terms, and Eq. (13) is the price. The problem of this paper is therefore a genuinely coupled two-channel problem in every angular momentum sector.

3.4 Symmetries

Lemma 2. *Let $\mu_n(l;\kappa)$ denote the eigenvalues of K_{I} in increasing order. Then (a) $\mu_n(-l-1;\kappa) = \mu_n(l;\kappa)$; (b) $\mu_n(l;\kappa) = \kappa\,\mu_n(l;1)$; (c) at $k = 0$ the spectrum of H in each channel is symmetric under $E \to -E$.*

Proof. (a) L_m depends on m^2 only, so $K_{\mathrm{I}}(-l-1)$ is $K_{\mathrm{I}}(l)$ with the two diagonal entries exchanged. With $U = \mathrm{i}\sigma_y$ one finds $UK_{\mathrm{I}}(l)U^{-1} = K_{\mathrm{I}}(-l-1)$. (b) The substitution $\rho \to \rho/\sqrt{\kappa}$ gives $K_{\mathrm{I}}(\kappa) = \kappa\,K_{\mathrm{I}}(1)$. (c) α^3 commutes with J_z, so it acts within a channel, and it anticommutes with $H(0)$. □

By (a), channels l and $-l-1$ (that is, $J_z = \pm(l+\frac{1}{2})$) have identical spectra.

4 Radial equation and the Kummer reduction

4.1 The reference case: a single oscillator

For the bosonic model, or for either diagonal entry of K_{I} alone, the radial equation is $(L_m + \kappa^2\rho^2 - \mu)f = 0$. Writing $f = \rho^m \mathrm{e}^{-\kappa\rho^2/2}\sum_j c_j\rho^{2j}$ gives the two-term recursion

$$(2j+2)(2j+2m+2)\,c_{j+1} = \left[2\kappa(2j+m+1) - \mu\right]c_j, \tag{15}$$

which terminates (so that f is normalizable) if and only if $\mu = 2\kappa(2n+m+1)$, $n = 0, 1, 2, \cdots$. This is Kummer's equation, and the termination argument is what fixes the spectrum in closed form.

4.2 Block I at $k = 0$

Set $k = 0$ in Eqs. (3a) and (3d). For $E \neq -\kappa\rho$ they give $h_2 = -\mathrm{i}\,(f_1' - lf_1/\rho)/(E+\kappa\rho)$, and eliminating h_2,

$$f_1'' + \left(\frac{1}{\rho} - \frac{\kappa}{E+\kappa\rho}\right)f_1' + \left(E^2 - \kappa^2\rho^2 - \frac{l^2}{\rho^2} + \frac{l\kappa}{\rho\,(E+\kappa\rho)}\right)f_1 = 0. \tag{16}$$

Its singular points are:

- $\rho = 0$, regular, with exponents $\pm l$;
- $\rho_* = -E/\kappa$, regular, with exponents 0 and 2. This point is *apparent*: (f_1, h_2) obey a first-order system whose coefficients are analytic at $\rho_* \neq 0$, so every solution is analytic there. It lies outside the physical half-line for $E > 0$ and on it for $E < 0$;
- $\rho = \infty$, irregular, with behavior $\mathrm{e}^{\pm\kappa\rho^2/2}$.

A single oscillator has only the first and the last. The second finite singular point arises when one component of the coupled system is eliminated, by dividing by $E + \kappa\rho$. We have not established whether a more general transformation of Eq. (16) to a known special-function equation exists.

An alternative is to eliminate h_2 from $K_{\mathrm{I}} u = \mu u$ directly. With $P_m = L_m + \kappa^2\rho^2 - \mu$ one finds $h_2 = \mathrm{i} P_l f_1/\kappa$ and the fourth-order equation

$$\left(P_{l+1} P_l - \kappa^2\right) f_1 = 0, \tag{17}$$

whose only finite singular point is $\rho = 0$ but whose order is four.

4.3 Series and recursion for $l = 0$

The regular solution of the first-order system with $l = 0$, $k = 0$ and $f_1(0) = 1$ has the expansion

$$f_1 = 1 - \frac{E^2}{4}\rho^2 - \frac{\kappa E}{18}\rho^3 + O(\rho^4), \qquad h_2 = \frac{\mathrm{i}E}{2}\rho + O(\rho^2) \tag{18}$$

. The odd power ρ^3 reflects the conical profile $M = \kappa\rho \propto |\boldsymbol{x}_\perp|$ of the mass at the axis. Stripping the Gaussian, $f_1 = \mathrm{e}^{-\kappa\rho^2/2} \sum_{n \geq 0} F_n \rho^n$, Eq. (16) with $l = 0$ is equivalent to

$$E(n+1)^2 F_{n+1} + \kappa\, n(n-1) F_n + E\,(E^2 - 2\kappa n) F_{n-1} + \kappa\left(E^2 - (2n-3)\kappa\right) F_{n-2} = 0, \tag{19}$$

with $F_{-1} = F_{-2} = 0$. For $E \neq 0$ the $n = 0$ equation gives $F_1 = 0$ and the recursion then determines every F_n from F_0; it reproduces the coefficients in Eq. (18) ($F_2 = -(E^2 - 2\kappa)/4$, $F_3 = -\kappa E/18$). Compared with Eq. (15), the recursion has four terms instead of two and contains the odd as well as the even powers, so the termination argument that fixes the oscillator spectrum does not apply directly to this expansion. We have not found a closed-form solution and make no claim about whether one exists.

Table 1 *Structure of the problem in the cases.*

System	Squared operator; radial equation	Spectrum
Boson, $m^2 \to m^2 + \kappa^2\rho^2$ [1]	$L_m + \kappa^2\rho^2$; Kummer, two-term recursion (15)	$\omega^2 = k^2 + m^2 + 2\kappa(2n + \lvert l\rvert + 1)$
Dirac oscillator, plane [11, 12]	non-minimal coupling (different operator structure); Kummer type [11, 12]	closed form, Laguerre type [11, 12]
Dirac, $M = m + g\lvert x\rvert$, 1D [7]	$p^2 + M^2 - M'\sigma_z$, constant σ_z; shifted oscillator on each half-line	transcendental matching condition [7]
Dirac, $M = \kappa\rho$ (this work)	$A + \kappa\Xi$, $[A, \Xi] \neq 0$ (Eq. (13)); second order, apparent singularity (Eq. (16)); four-term recursion (Eq. (19))	$E = \pm\sqrt{k^2 + \mu_n}$; no closed form found; Sec. 6

Note: The table was created by the student using LaTeX in Overleaf.

5 Spectrum: statements and proofs

Throughout, $l \in \mathbb{Z}$, $\kappa > 0$ and $k \in \mathbb{R}$. Three different operators appear, and their spectra differ: the squared transverse operator K_l (Sec. 5.2), the radial Dirac operator $H_{l,k}$ at fixed k (Sec. 5.3), and the Hamiltonian H of the full three-dimensional problem (Sec. 5.4). The first two have purely discrete spectra. The third does not: the mass profile confines the fermion in the transverse plane but not along the axis, and H has continuous subbands.

5.1 Radial operators and their domains

All L^2 spaces on $(0, \infty)$ carry the measure $\rho\, \mathrm{d}\rho$ unless stated otherwise. Let $\mathcal{H}_l = L^2((0,\infty), \rho\, \mathrm{d}\rho) \otimes \mathbb{C}^4$, with components (f_1, f_2, h_1, h_2), and let $\mathfrak{h}_k$ be the differential expression defined by the right-hand sides of Eqs. (3), so that $\mathfrak{h}_k u = Eu$. Then $\mathfrak{h}_k = \mathfrak{h}_0 + k\alpha^3$, where $\alpha^3(f_1, f_2, h_1, h_2) = (h_1, -h_2, f_1, -f_2)$ is the radial form of α^3. At $k = 0$ the expression splits into blocks acting on (f_1, h_2) and (h_1, f_2),

$$\mathfrak{h}_{\mathrm{I}} = \begin{pmatrix} \kappa\rho & -\mathrm{i}\big(\partial_\rho + \frac{l+1}{\rho}\big) \\ -\mathrm{i}\big(\partial_\rho - \frac{l}{\rho}\big) & -\kappa\rho \end{pmatrix}, \qquad \mathfrak{h}_{\mathrm{II}} = -\sigma_z\, \mathfrak{h}_{\mathrm{I}}\, \sigma_z \tag{20}$$

, so block II is unitarily equivalent to minus block I.

Lemma 3 (limit-point case; self-adjointness)**.** *For every $l \in \mathbb{Z}$ the expression $\mathfrak{h}_{\mathrm{I}}$ is limit-point at $\rho = 0$ and at $\rho = \infty$. Hence its closure H_{I} from $C_c^\infty((0,\infty))^2$ is self-adjoint, with domain $D(H_{\mathrm{I}}) = \{u \in L^2(\rho\, \mathrm{d}\rho)^2 :$ u locally absolutely continuous, $\mathfrak{h}_{\mathrm{I}} u \in L^2\}$. The same holds for $\mathfrak{h}_{\mathrm{II}}$, so $H_{l,0} := H_{\mathrm{I}} \oplus H_{\mathrm{II}}$ and $H_{l,k} := H_{l,0} + k\alpha^3$ are self-adjoint on $D(H_{l,0})$ for every real k. Moreover α^3 maps $D(H_{l,0})$ onto itself and anticommutes with $H_{l,0}$.*

Proof. The expression $\mathfrak{h}_{\mathrm{I}}$ is formally symmetric in $L^2(\rho\, \mathrm{d}\rho)^2$: its diagonal terms are real multiplication operators, and an integration by parts with the weight ρ shows $\int_0^\infty \rho\, \bar{a} \left(-\mathrm{i}(b' + \frac{l+1}{\rho} b)\right) \mathrm{d}\rho = \int_0^\infty \rho\, \overline{\left(-\mathrm{i}(a' - \frac{l}{\rho} a)\right)}\, b\, \mathrm{d}\rho$ for $a, b \in C_c^\infty((0,\infty))$, so the two off-diagonal operators are formal adjoints of each other. For such a first-order system Weyl's alternative holds [15]: an endpoint is limit-circle if, for one and then for every $z \in \mathbb{C}$, all solutions of $(\mathfrak{h}_{\mathrm{I}} - z)u = 0$ are square integrable near it, and limit-point otherwise. If both endpoints are limit-point, the deficiency indices of the minimal operator are $(0, 0)$; the minimal operator is then essentially self-adjoint and its closure is the maximal operator [15]. Because the dichotomy does not depend on z, it suffices to exhibit one solution that is not square integrable at each endpoint. We do so for an arbitrary $z \in \mathbb{C}$, in particular for the non-real values $z = \pm\mathrm{i}$ that enter the deficiency equations.

Endpoint $\rho = 0$. For $u = (f_1, h_2)^T$ the system $(\mathfrak{h}_{\mathrm{I}} - z)u = 0$ reads $u' = \rho^{-1}\mathrm{diag}(l, -l-1)\, u + B_z(\rho) u$ with $B_z(\rho) = \begin{pmatrix} 0 & \mathrm{i}(z + \kappa\rho) \\ \mathrm{i}(z - \kappa\rho) & 0 \end{pmatrix}$, polynomial in ρ. This is a regular singular point with exponents l and $-l-1$, which differ by the odd integer $2l + 1$. By the Frobenius theory for systems [16] there are solutions that behave to leading order as $\rho^l (1, 0)^T$ and $\rho^{-l-1}(0, 1)^T$ (the one with the smaller exponent may carry a logarithmic multiple of the other). With the measure $\rho\, \mathrm{d}\rho$ the first is square integrable near 0 if and only if

$l \geq 0$, the second if and only if $l \leq -1$. Hence for every l there is a solution that is not square integrable at 0 (the second for $l \geq 0$, the first for $l \leq -1$), and the endpoint is limit-point.

Endpoint $\rho = \infty$. Write $h_2 = \mathrm{i}g$. Then $(\mathfrak{h}_{\mathrm{I}} - z)u = 0$ becomes the first-order system

$$f_1' = \frac{l}{\rho} f_1 - (z + \kappa\rho)g, \quad g' = -\frac{l+1}{\rho} g + (z - \kappa\rho) f_1, \tag{21}$$

valid for every integer l. With $w = f_1 + g$, $v = f_1 - g$ and $a = l + \frac{1}{2}$,

$$\begin{aligned} w' &= -\Big(\kappa\rho + \frac{1}{2\rho}\Big)w + \Big(z + \frac{a}{\rho}\Big)v, \\ v' &= \Big(\kappa\rho - \frac{1}{2\rho}\Big)v + \Big(-z + \frac{a}{\rho}\Big)w \end{aligned} \tag{22}$$

. As long as $v \neq 0$, the ratio $R = w/v$ obeys

$$R' = -2\kappa\rho R + z + \frac{a}{\rho} + \Big(z - \frac{a}{\rho}\Big)R^2 \tag{23}$$

. The barrier estimate is easiest to state in dimensionless variables. Set $\xi = \sqrt{\kappa}\rho$ and $\zeta = z/\sqrt{\kappa}$, and in the remainder of this paragraph use primes for derivatives with respect to ξ. Equation (23) becomes

$$R' = -2\xi R + \zeta + \frac{a}{\xi} + \left(\zeta - \frac{a}{\xi}\right)R^2. \tag{24}$$

Let $q = |\zeta| + |a|$ and $\beta = q + 1$, and choose $\xi_1 > 1$ such that $\xi_1^2 > q\beta^2$. On the boundary $|R| = \beta/\xi$ for $\xi \geq \xi_1$, Eq. (24) yields the upper derivative bound

$$D_\xi^+|R| \leq -2\beta + q\left(1 + \frac{\beta^2}{\xi^2}\right) \leq -\beta < -\frac{\beta}{\xi^2} = \frac{\mathrm{d}}{\mathrm{d}\xi}(\beta/\xi),$$

where D_ξ^+ denotes the upper right derivative. Thus the region $|R| \leq \beta/\xi$ cannot be left once entered. Choose the solution initialized by $R(\xi_1) = 0$, corresponding to $w = 0$, $v = 1$ at $\rho_1 = \xi_1/\sqrt{\kappa}$; it stays in the region. In particular, v never vanishes there. In these variables,

$$\frac{v'}{v} = \xi - \frac{1}{2\xi} + \left(-\zeta + \frac{a}{\xi}\right)R = \xi + O(\xi^{-1}),$$

Taking real parts gives $\Re(v'/v) = \xi + O(\xi^{-1})$, so integration yields $|v| = \exp(\xi^2/2 + O(\log\xi))$. Since $|R| \leq \beta/\xi$, we have $R = O(\xi^{-1})$ and hence $|1 + R| \geq 1 - |R| \geq \frac{1}{2}$ for all sufficiently large ξ. Therefore $|f_1| = \frac{1}{2}|v||1 + R|$ is not square integrable with radial measure $\xi\,\mathrm{d}\xi$. The endpoint is therefore not limit-circle, hence limit-point. (A solution decaying like $\mathrm{e}^{-\xi^2/2}$ exists as well.)

The second block follows from Eq. (20). Since α^3 is a constant Hermitian matrix of norm one, $H_{l,k}$ is a bounded self-adjoint perturbation of $H_{l,0}$ and is self-adjoint on the same domain [14]. Finally $\mathfrak{h}_0\alpha^3 = -\alpha^3\mathfrak{h}_0$ as expressions, and the maximal domain is defined by square integrability, local absolute continuity and $\mathfrak{h}_0 u \in L^2$ alone, so α^3 preserves $D(H_{l,0})$ and anticommutes with $H_{l,0}$ there. □

Lemma 3 is the precise sense of the statement in Sec. 2 that the physical solutions are the regular ones: no boundary condition at the origin is needed, or allowed, beyond normalizability.

For $m \geq 0$ let A_m be the Friedrichs realization of $L_m + \kappa^2\rho^2$ in $L^2(\rho\,\mathrm{d}\rho)$, that is, the self-adjoint operator associated with the closure of the form $q_m[f] = \int_0^\infty \big(|f'|^2 + (m^2/\rho^2 + \kappa^2\rho^2)|f|^2\big)\rho\,\mathrm{d}\rho$ on $C_c^\infty((0,\infty))$ [14]. Its domain lies in the form domain, so f', mf/ρ and ρf are square integrable for $f \in D(A_m)$; moreover f, f' are locally absolutely continuous and $(L_m + \kappa^2\rho^2)f \in L^2$. For $m \geq 1$ the operator is the only self-adjoint realization (limit-point at 0); for $m = 0$ the Friedrichs choice selects the solution that is bounded at $\rho = 0$ and excludes $\ln\rho$, which has infinite q_0. Put $A = A_{|l|} \oplus A_{|l+1|}$ on $L^2(\rho\,\mathrm{d}\rho)^2$ and

$$\begin{aligned} K_{\mathrm{I}} &= A + \kappa\,\Xi_{\mathrm{I}} \;\text{ on } D(A), \\ K_{\mathrm{II}} &= \sigma_z K_{\mathrm{I}} \sigma_z, \qquad K_l = K_{\mathrm{I}} \oplus K_{\mathrm{II}}. \end{aligned} \tag{25}$$

K_{I} is self-adjoint because Ξ_{I} is bounded and Hermitian [14].

Lemma 4 (identification of the square)**.** *$D(K_l) \subset D(H_{l,0}^2)$ and $H_{l,0}^2 u = K_l u$ on $D(K_l)$. Consequently $H_{l,0}^2 = K_l$, and $H_{l,k}^2 = K_l + k^2$ with $D(H_{l,k}^2) = D(K_l)$ for every real k.*

Proof. It suffices to treat block I; block II follows by conjugation with σ_z and Eq. (20). Let $u = (f_1, h_2) \in D(A)$.

(i) *Regularity.* By the definition of A_m, f_1, h_2 and their first derivatives are locally absolutely continuous on $(0,\infty)$. Hence u is C^1, u' is locally absolutely continuous, and u'' exists almost everywhere and is locally square integrable.

(ii) *$\mathfrak{h}_{\mathrm{I}} u$ lies in L^2 and is locally absolutely continuous.* The functions f_1', h_2', $(l/\rho)f_1$, $((l+1)/\rho)h_2$, ρf_1 and ρh_2 are square integrable, because u lies in the form domain of A; hence $\mathfrak{h}_{\mathrm{I}} u \in L^2$. Moreover $\mathfrak{h}_{\mathrm{I}} u$ is built from u and u' with coefficients that are smooth on $(0,\infty)$, so it is locally absolutely continuous. By Lemma 3, $u \in D(H_{\mathrm{I}})$.

(iii) *$\mathfrak{h}_{\mathrm{I}}(\mathfrak{h}_{\mathrm{I}} u)$ lies in L^2.* Applying $\mathfrak{h}_{\mathrm{I}}$ twice involves u'', which exists almost everywhere by (i). Eq. (10) at $k = 0$ is a purely algebraic identity between expressions in u, u' and u'' (no integration by parts or boundary terms enter), so it holds almost everywhere for every such u and gives $\mathfrak{h}_{\mathrm{I}}(\mathfrak{h}_{\mathrm{I}} u) = K_{\mathrm{I}} u$. The right side is square integrable because $u \in D(A)$.

By (ii) and (iii), $\mathfrak{h}_{\mathrm{I}} u$ is square integrable and locally absolutely continuous, and $\mathfrak{h}_{\mathrm{I}}(\mathfrak{h}_{\mathrm{I}} u)$ is square integrable; that is, $\mathfrak{h}_{\mathrm{I}} u$ belongs to the maximal domain, which is $D(H_{\mathrm{I}})$ by Lemma 3. Therefore $u \in D(H_{\mathrm{I}}^2)$ and $H_{\mathrm{I}}^2 u = K_{\mathrm{I}} u$, so $K_{\mathrm{I}} \subset H_{\mathrm{I}}^2$. Both operators are self-adjoint; taking adjoints reverses the inclusion, $H_{\mathrm{I}}^2 \subset K_{\mathrm{I}}$, hence $H_{\mathrm{I}}^2 = K_{\mathrm{I}}$. For $k \neq 0$: for $u \in D(H_{l,0}) = D(H_{l,k})$ we have $\alpha^3 u \in D(H_{l,0})$ and $H_{l,0}\alpha^3 u = -\alpha^3 H_{l,0} u$ (Lemma 3), so $H_{l,k} u = H_{l,0} u + k\alpha^3 u \in D(H_{l,0})$ if and only if $H_{l,0} u \in D(H_{l,0})$. Hence $D(H_{l,k}^2) = D(H_{l,0}^2) = D(K_l)$ and, for such u, $H_{l,k}^2 u = H_{l,0}^2 u + k\{H_{l,0}, \alpha^3\}u + k^2 u = (K_l + k^2)u$. ☐

Remark (the logarithmic solution). The oscillator equation with $m = 0$ has the square-integrable solution $\ln\rho$ near the origin, so the operator A_0 requires a boundary condition. Lemma 4 shows that the operator equal to H^2 is the one with the Friedrichs condition $\lim_{\rho\to 0} \rho f'(\rho) = 0$ on the $m = 0$ component: any other self-adjoint realization A' yields $A' + \kappa\Xi_{\mathrm{I}} \neq K_{\mathrm{I}}$, which cannot be H_{I}^2. That condition follows from the Dirac domain itself, not from the choice of oscillator basis. Take $l = 0$, so that the $m = 0$ component is f_1, and $u \in D(H_{\mathrm{I}}^2)$: then $u \in D(H_{\mathrm{I}})$ and $\mathfrak{h}_{\mathrm{I}} u = (a, b) \in D(H_{\mathrm{I}})$. For any element (a, b) of $D(H_{\mathrm{I}})$ the first row of $\mathfrak{h}_{\mathrm{I}}(a, b) = (g_1, g_2) \in L^2$ gives $(\rho b)' = \mathrm{i}\rho(g_1 - \kappa\rho a)$; the right side is integrable at 0 by the Cauchy–Schwarz inequality with the weight ρ, so ρb has a limit at 0, and the limit vanishes because $b \in L^2(\rho\,\mathrm{d}\rho)$ excludes $b \sim c/\rho$ with $c \neq 0$. Hence $\rho b \to 0$ for every element of $D(H_{\mathrm{I}})$, and in particular $\rho h_2 \to 0$ for u itself. The second row of $\mathfrak{h}_{\mathrm{I}} u = (a, b)$ reads $b = -\kappa\rho h_2 - \mathrm{i} f_1'$, so $\rho f_1' = \mathrm{i}(\rho b + \kappa\rho^2 h_2) \to 0$. For $l = -1$ the $m = 0$ component is h_2, and the same argument with the two rows exchanged gives $\rho h_2' \to 0$. The Friedrichs condition is therefore a consequence of the domain of the Dirac operator.

5.2 The transverse operator: discreteness and bounds

On $L^2((0, \infty), \rho\,\mathrm{d}\rho)$ the operator A_m has the orthonormal eigenbasis

$$\psi_n^{(m)}(\rho) = \sqrt{\frac{2\kappa^{m+1} n!}{(n+m)!}}\,\rho^m L_n^{(m)}(\kappa\rho^2)\,\mathrm{e}^{-\kappa\rho^2/2}, \tag{26}$$

$$A_m \psi_n^{(m)} = 2\kappa(2n + m + 1)\,\psi_n^{(m)}, \tag{27}$$

$n = 0, 1, 2, \ldots$. The functions $\psi_n^{(m)}$ are bounded at the origin and have finite q_m, so they lie in $D(A_m)$; being a complete orthonormal set of eigenfunctions, they exhaust the spectrum. In block I the two components carry $m_1 = |l|$ and $m_2 = |l + 1|$, with $|m_1 - m_2| = 1$. Put $m_l = \min(m_1, m_2)$. The two arithmetic progressions $2\kappa(2n + m_1 + 1)$ and $2\kappa(2n + m_2 + 1)$, each with step 4κ, are offset by 2κ and interlace, so the eigenvalues of A in increasing order are simple and equal to

$$a_j = 2\kappa\,(j + m_l + 1), \qquad j = 0, 1, 2, \ldots \tag{28}$$

Theorem 1. *For every $l \in \mathbb{Z}$ and $\kappa > 0$, K_{I} is self-adjoint on $D(A)$ and has a purely discrete spectrum $\mu_0(l) \leq \mu_1(l) \leq \cdots \to \infty$ (counted with multiplicity); K_l has the same eigenvalues, each with twice the multiplicity. With $m_l = \min(|l|, |l + 1|)$ and for all $n = 0, 1, 2, \ldots$,*

$$\kappa\,(2n + 2m_l + 1) \leq \mu_n(l) \leq \kappa\,(2n + 2m_l + 3), \tag{29}$$

$$\mu_0(l) \leq 2\kappa\,(m_l + 1). \tag{30}$$

Proof. Ξ_{I} is a bounded Hermitian matrix with $\|\Xi_{\mathrm{I}}\| = 1$, so K_{I} is self-adjoint on $D(A)$ and its form domain equals that of A. The operator A has compact resolvent: its eigenvectors, the Laguerre functions of Eq. (26), form a complete orthonormal set with eigenvalues $a_j \to \infty$. Since $\kappa\Xi_{\mathrm{I}}$ is bounded, K_{I} has compact resolvent as well: for $|\mathrm{Im}\, z| > \kappa$ the resolvent $(K_{\mathrm{I}} - z)^{-1} = (A - z)^{-1}\left[1 + \kappa\Xi_{\mathrm{I}}(A - z)^{-1}\right]^{-1}$ is a compact operator times a bounded one. Hence the spectrum of K_{I} is purely discrete, consisting of eigenvalues $\mu_0 \leq \mu_1 \leq \ldots$ of finite multiplicity, and the j-th eigenvalue is given by the min–max principle [13]. For ψ in the form domain with $\|\psi\| = 1$, $\langle\psi, A\psi\rangle - \kappa \leq \langle\psi, K_{\mathrm{I}}\psi\rangle \leq \langle\psi, A\psi\rangle + \kappa$. By the min–max principle [13] the j-th eigenvalue obeys $a_j - \kappa \leq \mu_j \leq a_j + \kappa$, which is Eq. (29) by Eq. (28); in particular $\mu_j \to \infty$. For Eq. (30), take the trial vector with $\psi_0^{(m)}$ in the component that carries m and zero in the other. Then $\langle A\rangle = 2\kappa(m_l+1)$ by Eq. (27), while Ξ_{I} maps this vector into the other component, so $\langle \Xi_{\mathrm{I}}\rangle = 0$. The min–max principle gives $\mu_0(l) \le 2\kappa(m_l+1)$. $\square$

For $l=0$ the trial vector is the Gaussian $(\mathrm{e}^{-\kappa\rho^2/2}, 0, 0, 0)^T$ of the four-component problem; direct integration gives $\langle H(0)^2\rangle/\langle\psi|\psi\rangle = 2\kappa$.

5.3 The radial Dirac operator at fixed axial momentum

Proposition 1. *For every $l\in\mathbb{Z}$ and $k\in\mathbb{R}$ the radial Dirac operator $H_{l,k}$ has a purely discrete spectrum. It consists of*

$$E = \pm\sqrt{k^2+\mu_n(l)}, \qquad n = 0,1,2,\dots, \tag{31}$$

where $\mu_n(l)$ are the eigenvalues of K_{I}, and $+\sqrt{k^2+\mu_n}$ and $-\sqrt{k^2+\mu_n}$ occur with equal multiplicity. In particular $|E|\ge\sqrt{k^2+\kappa}$ in every channel.

Proof. By Lemma 4, $H_{l,k}^2 = K_l + k^2$, and $(H_{l,k}^2+1)^{-1} = (K_l+k^2+1)^{-1}$ is compact by Theorem 1; hence $H_{l,k}$ has compact resolvent and discrete spectrum. Let $V = \ker(K_l-\mu)\subset L^2(\rho\,\mathrm{d}\rho)^4$, which is finite dimensional and does not depend on k. By the spectral theorem, $V = \ker(H_{l,k}-\sqrt{\lambda})\oplus\ker(H_{l,k}+\sqrt{\lambda})$ with $\lambda = k^2+\mu>0$ ($\mu\ge\kappa$ by Theorem 1), so V is invariant under $H_{l,k}$ and the eigenvalues of $H_{l,k}$ on V are $\pm\sqrt{\lambda}$. It remains to show that the two signs occur equally often. The operator α^3 is bounded; it preserves $D(H_{l,0})$ and anticommutes with $H_{l,0}$ there (Lemma 3), hence commutes with $H_{l,0}^2 = K_l$ and maps V into itself. On V we may therefore write $H_{l,k}|_V = H_{l,0}|_V + k\,\alpha^3|_V$, a sum of Hermitian matrices on the finite-dimensional space V. The matrix $H_{l,0}|_V$ is invertible, with square μ, and anticommutes with $\alpha^3|_V$; hence $\alpha^3|_V = -H_{l,0}\,\alpha^3 H_{l,0}^{-1}$ and $\operatorname{tr}\alpha^3|_V = 0$. At $k=0$ the same anticommutation maps $\ker(H_{l,0}-\sqrt{\mu})$ onto $\ker(H_{l,0}+\sqrt{\mu})$, so $\operatorname{tr}H_{l,0}|_V = 0$. Therefore $\operatorname{tr}H_{l,k}|_V = 0$, i.e. $(n_+-n_-)\sqrt{\lambda} = 0$. $\square$

5.4 The three-dimensional Hamiltonian

The Fourier transform in z and the decomposition of the four components into the angular harmonics of Eq. (2) map $L^2(\mathbb{R}^3;\mathbb{C}^4)$ onto $\bigoplus_{l\in\mathbb{Z}}\int_{\mathbb{R}}^{\oplus}\mathcal{H}_l\,\mathrm{d}k$. We define the Hamiltonian H of the full problem as the operator that acts as $H_{l,k}$ on the fibers, $H\cong\bigoplus_l\int^{\oplus}H_{l,k}\,\mathrm{d}k$; this is self-adjoint because each fiber is self-adjoint (Lemma 3).

Corollary 1. *Let $E_* = \sqrt{\mu_0(0)}$. Then $\sqrt{\kappa}\le E_*\le\sqrt{2\kappa}$ and*

$$\operatorname{spec}H = (-\infty,-E_*]\cup[E_*,\infty). \tag{32}$$

H has no eigenvalues. For each l and n the dispersion relations $E = \pm\sqrt{k^2+\mu_n(l)}$ describe a continuous subband with threshold $\pm\sqrt{\mu_n(l)}$, and there is a gap $|E|<E_*$.

Proof. Theorem 1 gives $\sqrt{\kappa}\le E_*\le\sqrt{2\kappa}$. For $m_l\ge1$ the same theorem gives $\mu_0(l)\ge3\kappa>2\kappa\ge\mu_0(0)$, and $\mu_0(-1)=\mu_0(0)$ by Lemma 2(a); since $\mu_n(l)\ge\mu_0(l)$, every fiber satisfies $H_{l,k}^2 = K_l+k^2\ge E_*^2$.

Hence $H^2 \geq E_*^2$ and $\operatorname{spec} H \cap (-E_*, E_*) = \emptyset$. Conversely, the fibers $H_{l,k}$ depend continuously on k in the norm-resolvent sense (they differ by the bounded term $k\alpha^3$), and the spectrum of a decomposable self-adjoint operator is the essential closure of the union of the fiber spectra [13]. The continuous dispersion branch $k \mapsto \pm\sqrt{k^2 + \mu_0(0)}$ ensures that every energy in its range belongs to this essential closure; the two ranges fill $(-\infty, -E_*] \cup [E_*, \infty)$. If E were an eigenvalue, an eigenvector would have fibers satisfying $H_{l,k}\psi(k) = E\psi(k)$ for almost every k; for fixed E and l the equation $E^2 = k^2 + \mu_n(l)$ has at most two solutions k for each n and only finitely many n contribute, so this can hold only on a set of k of measure zero, and $\psi = 0$. □

Physical reading. The fermion is confined in the transverse plane and moves freely along the axis: each channel (l, n) is a relativistic subband with effective mass $\sqrt{\mu_n(l)}$. Although the mass vanishes on the axis, the transverse zero-point energy opens a gap of half-width $E_* = \sqrt{\mu_0(0)}$, between $\sqrt{\kappa}$ and $\sqrt{2\kappa}$.

6 Numerical spectrum

All numbers in this section are in units $\kappa = 1$ (energies in units of $\sqrt{\kappa}$, μ in units of κ); by Lemma 2(b) no generality is lost. The levels $\mu_n(l)$ are labelled by the position $n = 0, 1, 2, \ldots$ in the ordered spectrum of K_{I}, with both oscillator towers included.

6.1 Methods

We compute the radial spectrum using three formulations. Method A is two-sided matched shooting. Method B discretizes the first-order four-component Dirac operator in a Laguerre basis. Method C applies a Laguerre-basis Ritz approximation directly to the squared two-component operator. Methods B and C use the same family of basis functions but represent different operator formulations.

A. Matched shooting (block I, $k = 0$). With $h_2 = \mathrm{i}g$ and f_1, g real, the block-I system for $l \geq 0$ takes the exact same functional form as Eq. (21) but with $z \to E$:

$$f_1' = \frac{l}{\rho} f_1 - (E + \kappa\rho)\, g, \qquad g' = -\frac{l+1}{\rho}\, g + (E - \kappa\rho)\, f_1. \tag{33}$$

The regular solution starts at $\rho_0 = 10^{-6}$ with $f_1 = \rho_0^l$ and $g = E\rho_0^{l+1}/[2(l+1)]$. The solution that decays at large ρ has $g = f_1$ asymptotically, since $f_1 - g$ carries the growing exponential; it is integrated inward from $\rho_{\max} = \max(9.5, |E| + 5)$ starting from $(f_1, g) = (1, 1)$. Inward integration is the stable direction for this solution, so errors in the seed are damped rather than amplified. Both solutions are integrated to a matching radius $\rho_m = 2$ with the eighth-order DOP853 integrator (relative tolerance 10^{-13}). The eigenvalues are the zeros of $W(E)$, the sine of the angle between the two solution vectors at ρ_m. We scan E with step 0.02 over $0 < |E| \lesssim 9.6$ for $l = 0, 1$ and $0 < |E| \lesssim 4.4$ for $l = 2, 3$, bracket the sign changes of W, and polish each root with Brent's method. Negative l follows from Lemma 2(a). Integrating the regular solution outward alone would amplify round-off by $\mathrm{e}^{\kappa\rho^2/2}$; the two-sided matching avoids this.

B. First-order Galerkin (full four-component operator, any k). We expand (f_1, h_1) in the basis $\{\psi_n^{(|l|)}\}_{n<N}$ and (f_2, h_2) in $\{\psi_n^{(|l+1|)}\}_{n<N}$ of Eq. (26), which builds the correct behavior at the origin and the Gaussian decay into the basis. The matrix of Eqs. (3) is a Hermitian $4N \times 4N$ matrix; its elements are computed by Gauss–Legendre quadrature (4000 nodes on $[0, 24/\sqrt{\kappa}]$) with analytic basis derivatives.

C. Second-order Ritz (K_I). In the same basis K_I is $\begin{pmatrix} A_1 & \mathrm{i}\kappa S \\ -\mathrm{i}\kappa S^T & A_2 \end{pmatrix}$, where A_j is diagonal with entries $2\kappa(2n + m_j + 1)$ and $S_{nn'} = \langle \psi_n^{(m_1)} | \psi_{n'}^{(m_2)} \rangle$. By the min–max principle the eigenvalues of this matrix are upper bounds to the exact μ_n.

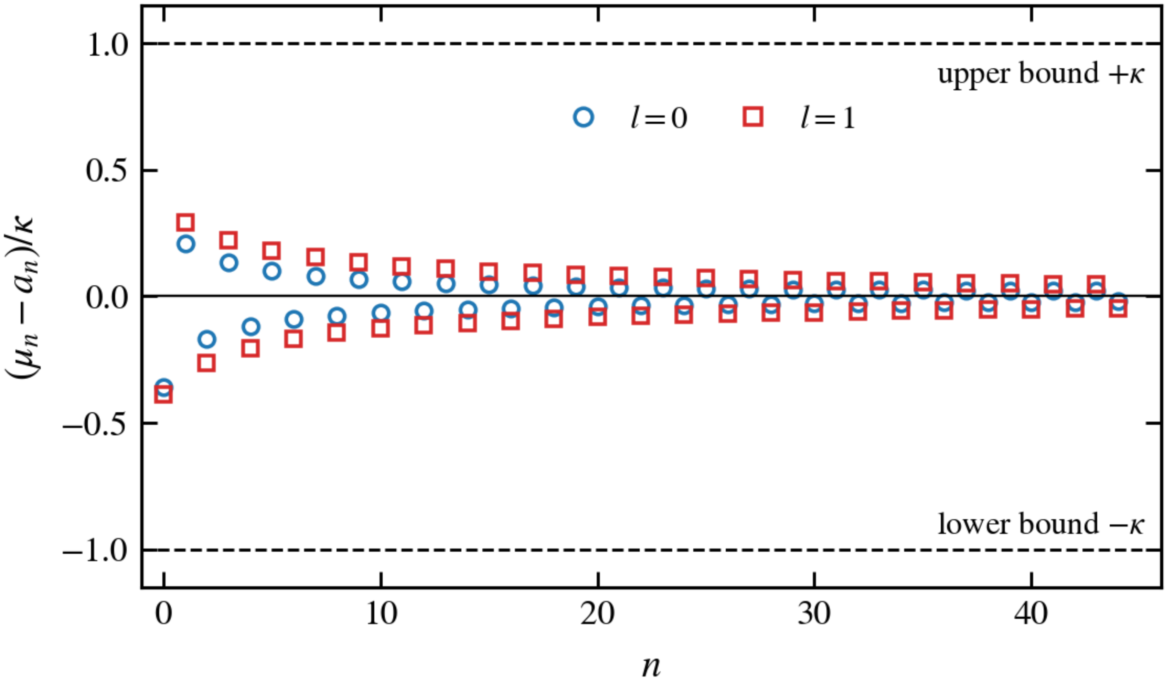
6.2 Basis-size convergence

Table 4 reports the largest absolute difference from the matched-shooting reference among the four lowest $l = 0$ levels for each Laguerre-basis formulation. The first-order matrix eigenvalues are squared before comparison with μ_n. The second-order Ritz eigenvalues are upper approximations to the exact levels by the min–max principle, up to the finite-quadrature error in the overlap matrix. For each method, a least-squares fit of $\log(\text{error})$ against $\log N$ over $N = 40, 60, 80, 100$ gives an observed convergence rate of approximately $N^{-3.1}$ for the first-order formulation and $N^{-2.0}$ for the second-order formulation. These exponents describe the finite range of basis sizes shown in Table 4; they are not asserted as asymptotic theorems. At $N = 100$, the largest differences over the four levels are 8.0×10^{-8} for the first-order formulation and 2.5×10^{-6} for the second-order formulation.

6.3 Results

Figure 1

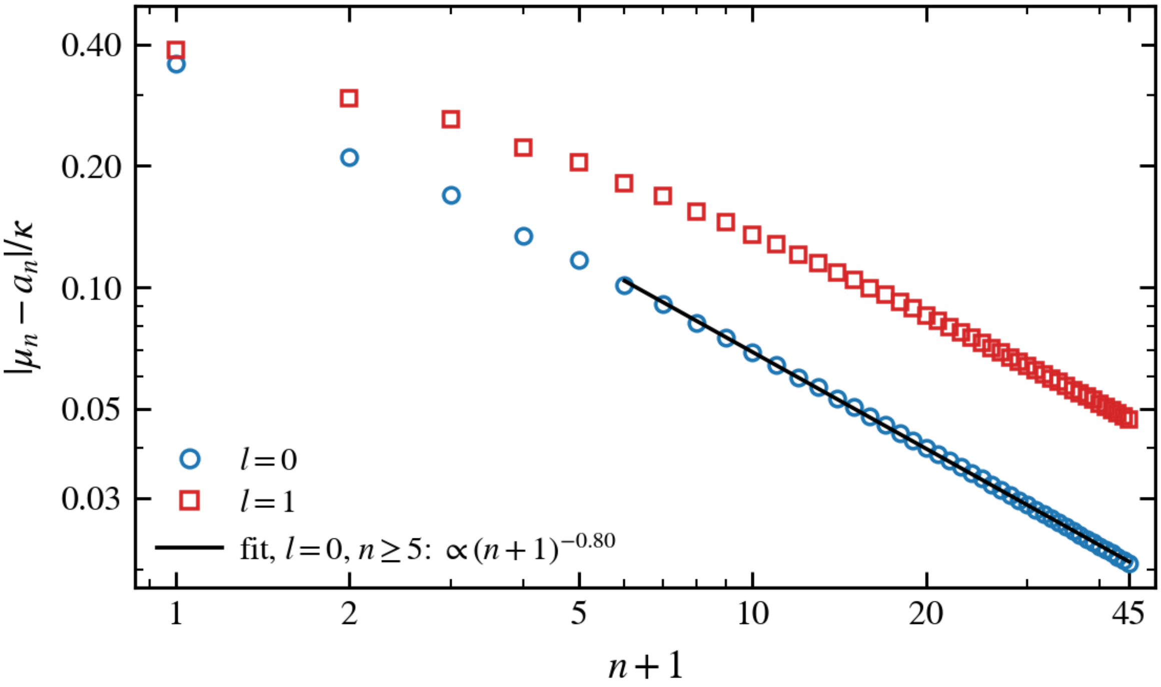
Signed Deviations From the Oscillator Reference Ladder



Note. The plotted quantity is $(\mu_n - a_n)/\kappa$, where $a_n = 2\kappa(n + m_l + 1)$, for $l = 0$ (blue open circles) and $l = 1$ (red open squares). The dashed horizontal lines show the proven normalized bounds ±1, equivalent to $|\mu_n - a_n| \leq \kappa$. The observed sign alternation and decreasing magnitude are numerical observations, not theorems. The graphic was produced by the student by outputting a graph using python on Google Collab, 2026.

Figure 2

Magnitude of the Spectral Deviations



Note. The absolute normalized deviations $|\mu_n - a_n|/\kappa$ are plotted against $n + 1$ on logarithmic axes for $l = 0$ (blue open circles) and $l = 1$ (red open squares). The black line is a least-squares power-law fit to the $l = 0$ data for $n \geq 5$, with a fitted exponent of approximately -0.80. This fit describes the computed finite-range data and is not a claim about the asymptotic behavior. The graphic was produced by the student by outputting a graph using python on Google Collab, 2026.

Table 2 *Channel $l = 0$, $\kappa = 1$: the lowest levels μ_n, the deviation $\delta_n = \mu_n - 2(n+1)$ from the oscillator ladder, and the energies at $k = 0$ and $k = \frac{1}{2}$. Values are from Method A (about 10^{-12} absolute accuracy); $E_n(k) = \sqrt{k^2 + \mu_n}$ is exact by Proposition 1.*

n	μ_n	δ_n	$E_n(k{=}0)$	$E_n(k{=}\frac{1}{2})$
0	1.64113736	−0.358863	1.28106883	1.37518630
1	4.20980088	0.209801	2.05177993	2.11182407
2	5.83058791	−0.169412	2.41466103	2.46588481
3	8.13457649	0.134576	2.85211789	2.89561332
4	9.88291245	−0.117088	3.14370998	3.18322359
5	12.10098556	0.100986	3.47864709	3.51439690
6	13.90893778	−0.091062	3.72946883	3.76283640
7	16.08164477	0.081645	4.01019261	4.04124297
8	17.92481660	−0.075183	4.23377097	4.26319324
9	20.06894869	0.068949	4.47983802	4.50765446
10	21.93562457	−0.064375	4.68354829	4.71016184
11	24.05991746	0.059917	4.90509097	4.93050884

Note: The table was created by the student using LaTeX in Overleaf.

Table 3 *Lowest three levels μ_n in the channels $l = 0, \dots, 3$ ($\kappa = 1$; $l \to -l - 1$ gives the same values). The last column is the proven window for $\mu_0(l)$, $[\kappa(2m_l + 1),\ 2\kappa(m_l + 1)]$ from Eqs. (29) and (30).*

l	μ_0	μ_1	μ_2	window for μ_0
0	1.641137	4.209801	5.830588	[1, 2]
1	3.613183	6.293460	7.738277	[3, 4]
2	5.603815	8.328237	9.696552	[5, 6]
3	7.599191	10.347396	11.672764	[7, 8]

Note: The table was created by the student using LaTeX in Overleaf.

Table 4 *Convergence of the Laguerre-basis formulations* ($l = 0$, $\kappa = 1$)*: largest error over the four lowest levels, first order* $\max_n |E_{n,N}^2 - \mu_n^{\mathrm{ref}}|$ *and second order* $\max_n |\mu_{n,N} - \mu_n^{\mathrm{ref}}|$. *The exponents quoted in the text are least-squares fits of* $\log(\mathrm{error})$ *against* $\log N$ *over* $N = 40$–100*;* $N = 20$ *is shown but not fitted.*

N	first order	second order
20	1.18×10^{-5}	6.19×10^{-5}
40	1.34×10^{-6}	1.56×10^{-5}
60	3.82×10^{-7}	6.93×10^{-6}
80	1.58×10^{-7}	3.90×10^{-6}
100	7.96×10^{-8}	2.50×10^{-6}

Note: The table was created by the student using LaTeX in Overleaf.

Table 2 lists the lowest levels of the channel $l = 0$. The ground state has $E_0 = 1.2810688331\sqrt{\kappa}$ at $k = 0$, above the proven lower bound $\sqrt{\kappa}$ and below $\sqrt{2\kappa}$. Table 3 shows the channel dependence. All 45 levels computed for $l = 0$, all 45 for $l = 1$, and all levels computed for $l = 2$ (7) and $l = 3$ (6) lie inside the proven windows.

Proven and observed. Equation (30) proves $\delta_0 \leq 0$ in every channel. For $l = 0, 1, 2, 3$, the computed values are

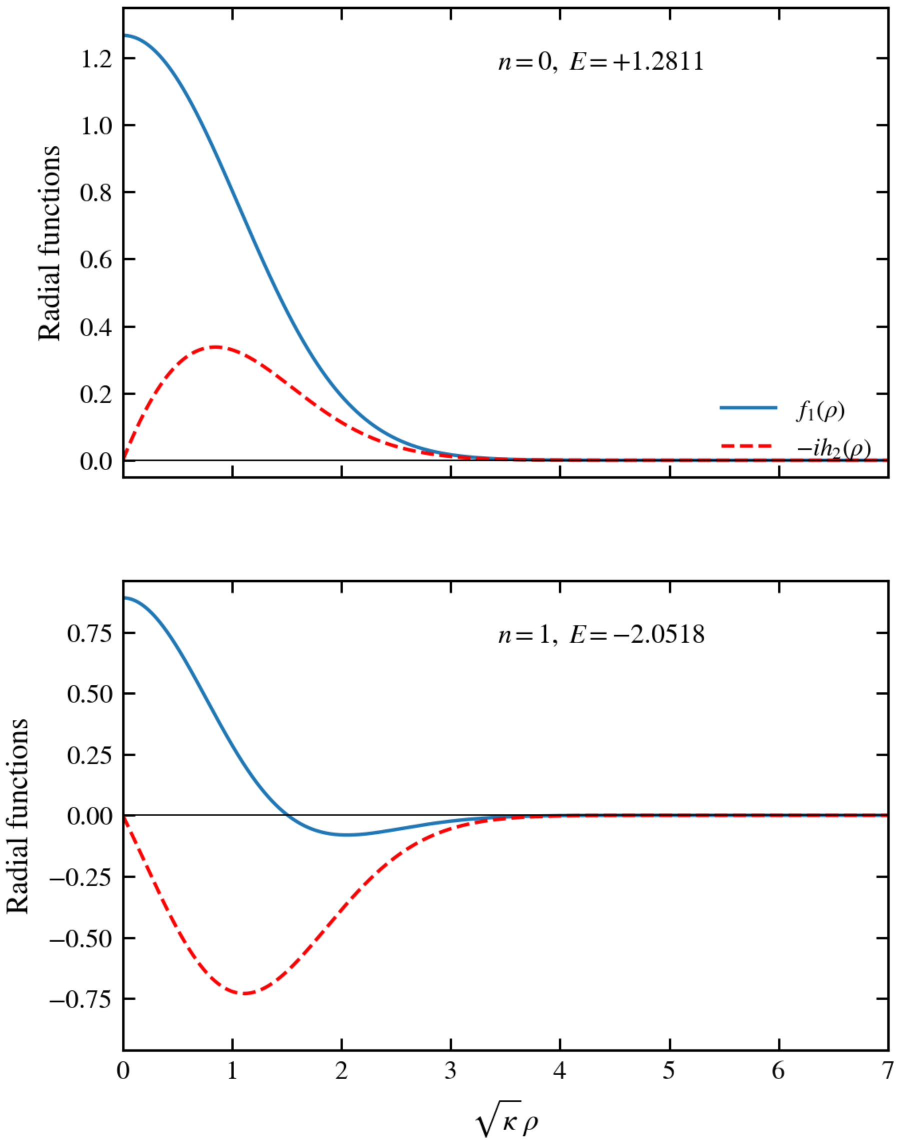
$$(\delta_0(0), \delta_0(1), \delta_0(2), \delta_0(3)) = (-0.3589, -0.3868, -0.3962, -0.4008).$$

Thus $|\delta_0|$ grows slowly with l over these four channels. Beyond this, the following are *observations* that the bounds do not imply: (a) the deviations alternate in sign (Figure 1), $\operatorname{sgn}\delta_n = (-1)^{n+1}$, in all four channels and for every level computed; (b) $|\delta_n|$ decreases monotonically over all computed levels in all four channels (Figure 2), roughly as $(n+1)^{-0.80}$ for $l = 0$, $n \geq 5$. This power law describes the data and is not an asymptotic claim. We do not have proofs of (a) or (b).

Structure of the states. The computed levels alternate between states dominated by the two uncoupled oscillator components: even n are associated predominantly with the first component, and odd n with the second. This component assignment is a numerical pattern, not a theorem. For the two states shown in Figure 3, the component h_2 carries only 12.9% of the norm for $n = 0$, whereas it carries 81.2% for $n = 1$. Both radial functions are regular at the origin, with $f_1(0) \neq 0$ and $h_2 \propto \rho$, as in Eq. (18).

Figure 3

Radial Wavefunctions for the Two Lowest Squared-Spectrum Levels



Note. The radial functions are shown for $l = 0$, $k = 0$, and $\kappa = 1$. They are reconstructed from the signed block-I roots and normalized so that $\int_0^\infty (|f_1|^2 + |h_2|^2)\rho\, \mathrm{d}\rho = 1$, with $h_2 = \mathrm{i}g$ and g real. The upper panel shows the $n = 0$ state at $E = +1.2810688331$; the lower panel shows the $n = 1$ state at $E = -2.0517799301$. Block II supplies the partner with the opposite signed energy. The graphic was produced by the student by outputting a graph using python on Google Collab, 2026.

7 Discussion and conclusions

What the structure says. The proven bounds have a simple reading. The oscillator part A of K_{I} is the direct sum of two bosonic problems of the type solved in Ref. [1], with angular momenta $|l|$ and $|l + 1|$, and its levels interlace into the single ladder $a_j = 2\kappa(j + m_l + 1)$, where $m_l = \min(|l|, |l+1|)$ (Eq. (28)). The coupling $\kappa\Xi_{\mathrm{I}}$ has norm κ, so it can move each level by at most κ. To within $\pm\kappa$ per level, the fermionic spectrum is therefore the union of two bosonic spectra, and the numerics show that the actual shift is smaller than the bound: at most $0.3589\,\kappa$ in magnitude for $l = 0$ (Table 2, Figures 1 and 2). The coupling does not preserve the direct reduction to Kummer's equation available for the bosonic and Dirac-oscillator problems. In every channel the centrifugal terms of the two components differ by $(2l+1)/\rho^2$, Eq. (13), so no constant change of component basis decouples them, and one is left with the coupled problem of Sec. 4.

Limitations. (i) The mass vanishes on the axis; a bare mass, for instance $M = \sqrt{m^2 + \kappa^2\rho^2}$, would make $M'(\rho)$ non-constant and the squared operator no longer of the form $A + \kappa\Xi$ with constant κ. (ii) The profile $M = \kappa\rho$ is conical at the axis. (iii) The numerics explicitly cover $l = 0, 1, 2, 3$, with the corresponding negative-channel spectra obtained by symmetry, and a finite number of levels; the statements about alternation and decay are observations, not theorems. (iv) No closed form is found, and none is claimed to be impossible.

Conclusions. A Dirac fermion with a scalar mass $M = \kappa\rho$ has an exactly separable axial momentum and a transverse operator that squares to two coupled oscillators in every angular-momentum channel. For each axial momentum the radial Dirac operator is self-adjoint with a purely discrete spectrum $E = \pm\sqrt{k^2 + \mu_n(l)}$, where $m_l = \min(|l|, |l+1|)$ and $\kappa(2n + 2m_l + 1) \le \mu_n(l) \le \kappa(2n + 2m_l + 3)$; the full Hamiltonian has continuous subbands with a gap $|E| < E_*$, since the mass profile confines the fermion transversely but not along the axis. The direct reduction to Kummer's equation does not apply. The spectrum is nevertheless accessible to high-accuracy numerics, which we have provided for the lowest levels of four channels.

Computational assistance

The initial code for the numerical spectrum calculations was created by ChatGPT (OpenAI) for which the code was audited and reprogrammed to accurately model the spectrum.

A Derivation of the radial equations

In the Dirac representation $H = \begin{pmatrix} M & \boldsymbol{\sigma}\cdot\boldsymbol{p} \\ \boldsymbol{\sigma}\cdot\boldsymbol{p} & -M \end{pmatrix}$ with $\boldsymbol{\sigma}\cdot\boldsymbol{p} = \sigma_z k + \begin{pmatrix} 0 & p_x - \mathrm{i}p_y \\ p_x + \mathrm{i}p_y & 0 \end{pmatrix}$ at fixed $p_z = k$. In polar coordinates $p_x \pm \mathrm{i}p_y = -\mathrm{i}\,\mathrm{e}^{\pm\mathrm{i}\varphi}(\partial_\rho \pm \frac{\mathrm{i}}{\rho}\partial_\varphi)$, so that

$$(p_x + \mathrm{i}p_y)\big(\mathrm{e}^{\mathrm{i}l\varphi}u(\rho)\big) = -\mathrm{i}\,\mathrm{e}^{\mathrm{i}(l+1)\varphi}\Big(u' - \frac{l}{\rho}u\Big), \tag{34}$$

$$(p_x - \mathrm{i}p_y)\big(\mathrm{e}^{\mathrm{i}(l+1)\varphi}u(\rho)\big) = -\mathrm{i}\,\mathrm{e}^{\mathrm{i}l\varphi}\Big(u' + \frac{l+1}{\rho}u\Big). \tag{35}$$

Inserting the ansatz (2) into $H\psi = E\psi$ and reading off the four rows gives

$$\begin{aligned}
&\text{row 1:}\quad \kappa\rho f_1 + k\, h_1 - \mathrm{i}\big(h_2' + \tfrac{l+1}{\rho} h_2\big) = E f_1,\\
&\text{row 2:}\quad \kappa\rho f_2 - k\, h_2 - \mathrm{i}\big(h_1' - \tfrac{l}{\rho} h_1\big) = E f_2,\\
&\text{row 3:}\quad -\kappa\rho h_1 + k\, f_1 - \mathrm{i}\big(f_2' + \tfrac{l+1}{\rho} f_2\big) = E h_1,\\
&\text{row 4:}\quad -\kappa\rho h_2 - k\, f_2 - \mathrm{i}\big(f_1' - \tfrac{l}{\rho} f_1\big) = E h_2,
\end{aligned}$$

which are Eqs. (3a)–(3d).

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